

Sleep Apnea Detection From Single-Lead ECG: A Comprehensive Analysis of Machine Learning and Deep Learning Algorithms

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Abstract—Sleep apnea is a common sleep breathing disorder (SBD) in which patients suffer from stopping or decreasing airflow to the lungs for more than 10 sec. Accurate detection of sleep apnea episodes is an important step in devising appropriate therapies and management strategies. This article provides a comprehensive analysis of machine learning and deep learning algorithms on 70 recordings of the PhysioNet ECG Sleep Apnea v1.0.0 dataset. First, electrocardiogram signals were pre-processed and segmented and then machine learning and deep learning methods were applied for sleep apnea detection. Among conventional machine learning algorithms, linear and quadratic discriminate analyses, logistic regression, Gaussian naïve Bayes, Gaussian process, support-vector machines, k -nearest neighbor, decision tree, extra tree, random forest, AdaBoost, gradient boosting, multi-layer perceptron, and majority voting were implemented. Among deep algorithms, convolutional networks (Alex-Net, VGG16, VGG19, ZF-Net), recurrent networks (LSTM, bidirectional LSTM, gated recurrent unit), and hybrid convolutional-recurrent networks were implemented. All networks were similarly modified to handle our biosignal processing task. The available data were divided into a training set to adjust the model parameters, a validation set to adjust hyperparameters, avoid overfitting, and improve the generalizability of the models, and a test set to evaluate the generalizability of the models on unseen data. This procedure was then repeated in a fivefold cross-validation scheme so that all the recordings were once located in the test set. It was found that the best detection performance is achieved by hybrid deep models where the best accuracy, sensitivity, and specificity were 88.13%, 84.26%, and 92.27%, respectively. This study provides valuable information on how different machine learning and deep learning algorithms perform in the detection of sleep apnea and can further be extended toward the detection of other sleep events. Our developed algorithms are publicly available on GitHub.

Index Terms—Deep learning, detection, electrocardiogram (ECG), machine learning, sleep apnea.

I. INTRODUCTION

SLEEP accounts for about one-third of the human life span and plays an important role in health and quality of life. Two main stages of sleep are called rapid eye movement (REM) and non-REM (NREM) [3]. REM sleep compared with NREM sleep is associated with increased sympathetic activity, cardiovascular instability, and hemodynamic variations. During NREM, oxygen consumption and heart rate decrease as well as the blood pressure, while in the REM stage, blood pressure and heart rate surge. Sleep apnea may occur during any stage of sleep, but it is most dominant during the REM. This is mainly because the muscles in the upper airway are further relaxed during the REM stage of sleep.

The rate of people suffering from sleep disorders is increasing [6]. In the United States, it is estimated that about 70 million adults have sleep disorders [8]. The rate of morbidity and mortality increases in patients suffering from sleep disorders [10]. Insomnia, sleep breathing disorders (SBDs), central disorders of hypersomnolence, circadian rhythm sleep-wake disorders, parasomnias, and movement disorders are the main types of sleep disorders. Among different sleep disorders, SBDs are more common. SBDs are divided into obstructive sleep apnea (OSA), central sleep apnea (CSA), hypoventilation in sleep, and hypoxia disorders. Upper airway obstruction leads to OSA, whereas CSA occurs when the brain cannot send proper signals into muscles responsible for breathing. It is estimated that one billion people in the world are suffering from OSA [12]. CSA occurs less than other types of apneas. CSA mostly occurs in the NREM cycle, while OSA occurs in both REM and NREM cycles but the severity of OSA in the REM cycle is higher [14].

To prevent and manage cardiovascular, mental, and other health issues related to sleep apnea, accurate, affordable, and portable technologies are required that can detect and monitor sleep events.

There has been considerable research on the development of automatic algorithms for the detection of sleep apnea from a variety of physiological signals [15]–[22] among which ECG has shown the most promising results in terms of both convenience and precision [2], [4], [5],

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[7], [9], [11], [13], [23]. Apnea detection based on a single-lead electrocardiogram (ECG) provides the opportunity for the design and development of wearable technologies based on a single sensor.

Conventional machine learning algorithms are among the first automatic algorithms used for the detection of apnea events. Given the complexity of physiological signals and the limited feature extraction capability of conventional machine learning algorithms, the research focus has recently been shifted toward the more complex deep learning models. Thanks to the advancements in computational power, deep learning algorithms have surpassed their conventional machine learning counterparts by having the capability of automatic extraction of most representative features. Given the variety of the algorithms designed for the detection of sleep apnea, the different datasets and physiological signals used for training them, and the various validation strategies as well as performance metrics used for evaluation of the algorithms, it is hard to compare the algorithms' performance.

In this article, we provide a fair and unbiased comparison between different conventional machine learning and deep learning algorithms in the detection of sleep apnea occurrence from a single-lead ECG. All the experiments are performed on the same dataset and under the same setting to be able to properly evaluate and compare the performances of different algorithms. Unlike most studies that tune their model hyperparameters based on the same data used for final evaluation, we used three sets of data: a training set to train the model parameters, a validation set to find the model optimum hyperparameters, and a test set to evaluate the generalizability of the developed models on unseen data. This article is an extended version of our work presented in the 16th edition of IEEE International Symposium on Medical Measurements and Applications [1], where the performance of a few deep learning methods was analyzed for the detection of sleep apnea. In this article, we have significantly extended our comparisons by studying the performance of 14 different conventional machine learning methods and 19 different deep learning methods. A detailed comparison between different models as well as eight other state-of-the-art techniques is provided. We also performed a feature importance analysis and demonstrated which ECG features are most effective for the detection of apnea episodes. We have made our algorithms publicly available on GitHub at <https://github.com/mahsaabahrami/sleep-apnea>.

II. BACKGROUND

Accurate detection of sleep apnea is an important step for the management of apnea side effects. Questionnaires, image processing techniques, and signal analysis methods are the main approaches for sleep apnea monitoring. Questionnaires are low-cost methods for the identification of patients with sleep disorders [24]. Medical image analysis is a capable method for analyzing sleep apnea especially in severe sleep apnea [25]. Medical image analysis also provides a unique opportunity for screening anatomical changes during sleep apnea. Polysomnography (PSG) is a gold standard for sleep

apnea monitoring [26]. PSG consists of various biological signals, such as ECG, electroencephalogram, airflow pulse oximetry, arterial blood oxygen saturation, and nasal flow. However, none of these techniques can be performed in non-specialist settings, such as in home. Wearable and non-contact sleep technologies are required that can unobtrusively monitor and manage sleep without the need for direct expert presence [27], [28]. Different physiological signals have been analyzed for the detection of sleep apnea among which ECG provides a better possibility for implementation in wearable monitors.

Conventional machine learning and novel deep learning are two main automatic approaches for sleep apnea monitoring from physiological signals. The performance of conventional machine learning algorithms relies on expert-defined extracted features. In [15], ECG signals were decomposed to frequency bands by Fourier transform, and statistical features were extracted. A support vector machine (SVM) with Gaussian kernel function was then used to detect sleep apnea. In [2], the performance of different machine learning methods was compared. For this purpose, after extracting different time-domain features of heart rate variability (HRV) and ECG-derived respiration, artificial neural network (ANN), SVM, linear discriminate analysis (LDA), partial least-square regression, and augmented naïve Bayes regression were applied for the detection of sleep apnea events. The best accuracy and sensitivity were achieved by ANN; however, the specificity of augmented naïve Bayes regression was higher than other methods.

Recently, deep learning methods have revolutionized sleep apnea detection. Despite conventional machine learning approaches, deep learning methods rely on automatic feature extraction. Deep recurrent neural networks (DRNNs) and convolutional neural networks (CNNs) are two main types of deep learning models that have been used for sleep apnea monitoring. In [4], continuous wavelet transformation and Alex-Net were used for feature extraction and then different machine learning methods were used for the classification of sleep apnea episodes from single-lead ECG signals. It was shown that extracting time-frequency features improves the performance of sleep apnea detection. In [5], it was shown that the accuracy of sleep apnea detection can be improved by a two-layer CNN. In [7], a long short-term memory (LSTM) was used for OSA detection from single-lead ECG that provided an accurate and real-time method for sleep apnea detection. In [9], unsupervised feature learning from ECG signal was done by a frequential stacked auto-encoder, and a hidden Markov model was used for classification. In [11], a multiscale dilation attention convolution module was used and was shown that the combination of a multiscale attention convolutional module and a weighted-loss time-dependent classifier improves the performance of sleep apnea detection. In [1], we performed a comparison between a few deep networks and found that a hybrid CNN-LSTM achieves superior results.

In [29], a real-time sleep apnea detection method based on motion detection was proposed by using quadrature microwave Doppler radar. In [30], a wearable device based on single-lead ECG was proposed for improving the performance of home

monitoring devices. In [31], a phase-modulated continuous radar system was proposed that could detect stop breathing events under predefined situations. Sleep wearable devices that can automatically monitor and manage sleep and health issues will form an important component of future health-care systems. Therefore, further research is required on the evaluation of different artificial intelligence models under the same settings on the same datasets to discover their potentials and limitations to choose the appropriate algorithms that can perform well in smart wearables.

III. EXPERIMENTAL SETUP

A. Dataset

In this article, PhysioNet Apnea-ECG Database v1.0.0 [32] was used to build our models and to compare their performances. The database contains 70 recordings from 32 individuals (seven females, age: 44 ± 11 , apnea-hypopnea index, AHI: 24 ± 25 , height: 175 ± 6 , weight: 86 ± 22) divided into four groups: A, B, C, and X. Thirteen individuals in this dataset have $AHI \leq 5$ and are categorized as normal, 13 individuals have $AHI > 30$, and are categorized in the severe sleep apnea class, and six individuals have $5 < AHI < 30$ and are categorized in the mild and moderate apnea class. The length of the 70 recordings is 8.2 ± 0.52 h. ECG signals are digitized at 100 Hz. The annotation of the presence of sleep apnea is provided for every 1 min by a sleep expert.

B. Preprocessing

ECG signals were segmented into 1-min intervals. To extract the R - R Intervals from the ECG signal, the Hamilton R -peak detection method based on an open-source code was employed [33]. A median filter proposed by Chen *et al.* [34] was used for removing physiologically uninterpretable points.

The extracted R - R Intervals were then presented to the developed machine learning algorithms after appropriate processing as is discussed in the Section IV.

In addition to the R - R Intervals, the amplitudes of R -peaks were also extracted and presented to the deep models. Cubic interpolation at 3 Hz was applied to sample R - R intervals and R -peak amplitudes at an equal rate. Finally, the interpolated R - R Intervals and R -peak amplitudes were fed to the deep models. To capture the time pattern of the input data when using DRNNs, the input data were segmented into n segments of $60/n$ s and n -cell DRNNs were designed to learn from the input data (here, n was set to 2).

IV. CONVENTIONAL MACHINE LEARNING

Feature extraction, feature selection, and classification are the main steps in conventional machine learning methods. Feature engineering plays a vital role in boosting the performance of these machine learning algorithms. Here, we developed a unified feature engineering framework for the conventional machine learning algorithms (shown in Fig. 1). First, ECG signals were pre-processed. Then time, frequency, and nonlinear features of the ECG signal were extracted. In addition to this, principal component analysis (PCA) was applied for

dimension reduction, coping with the curse of dimensionality, and extracting the most robust features. Finally, the ECG signal was classified into apnea and normal episodes.

A. Feature Extraction

The most important features of ECG for apnea detection are HRV parameters [35]. In this article, a comprehensive set of time-domain, frequency-domain, and nonlinear features of HRV were extracted and presented to different machine learning models for sleep apnea detection.

1) *Time-Domain Features*: Time-domain features of HRV are mainly divided into statistical, long-term, and short-term features [36]. Minimum, range, median, mean, standard deviation (SDNN), skewness (third momentum), and kurtosis (fourth momentum) were the main statistical features extracted from R - R intervals. The standard deviation of successive differences between adjacent R - R Intervals (SDSD) was the main long-term extracted feature of HRV. Short-term features of HRV consisted of the number of R - R intervals that differ from each other by more than 50 ms (NN50), the number of R - R intervals that differ from each other by more than 20 ms (NN20), percentage of successive R - R differences greater than 50 ms (pNN50), and the percentage of successive R - R differences greater than 20 ms (pNN20). In addition, the root mean square of successive differences between normal heartbeats (rMSSD) was extracted from the HRV signal. In total, 13 features were extracted from each ECG segment in the time domain.

2) *Frequency-Domain Features*: HRV has three predominant frequency bands [36]. Very low-frequency (VLF) band, that is, frequencies between 0.0033 and 0.04 Hz, exhibits the thermoregulation mechanisms. Low-frequency (LF) band, that is, frequencies between 0.04 and 0.15 Hz, exhibits mostly the sympathetic activity, and the high-frequency (HF) band, that is, frequencies between 0.15 and 0.4 Hz, exhibits parasympathetic activities.

Total power, VLF power, LF power, HF power, LF to HF ratio, normalized HF power (HFnu), and normalized LF power (LFnu) were extracted. In total, seven features were extracted from each ECG segment in the frequency domain.

3) *Nonlinear Features*: Nonlinear features are mainly used for modeling and analyzing ECG patterns before, during, and after sleep apnea. Poincaré plot features including SD1 and SD2, and their ratio (SD2/SD1) [37], cardiovagal index (CVI) [38], cardio sympathetic index (CSI) [38], and modified CVI [38] were extracted from the ECG signal. Permutation entropy [39] explaining the complexity of the HRV time series was also calculated. In total, seven nonlinear HRV features were extracted.

In total, 27 HRV features were extracted.

B. Dimension Reduction

Many irrelevant features lead to the curse of dimensionality problem and have a detrimental impact on the performance of machine learning methods. To cope with these problems, feature selection techniques are applied before classification. Feature selection provides a unique opportunity for detecting

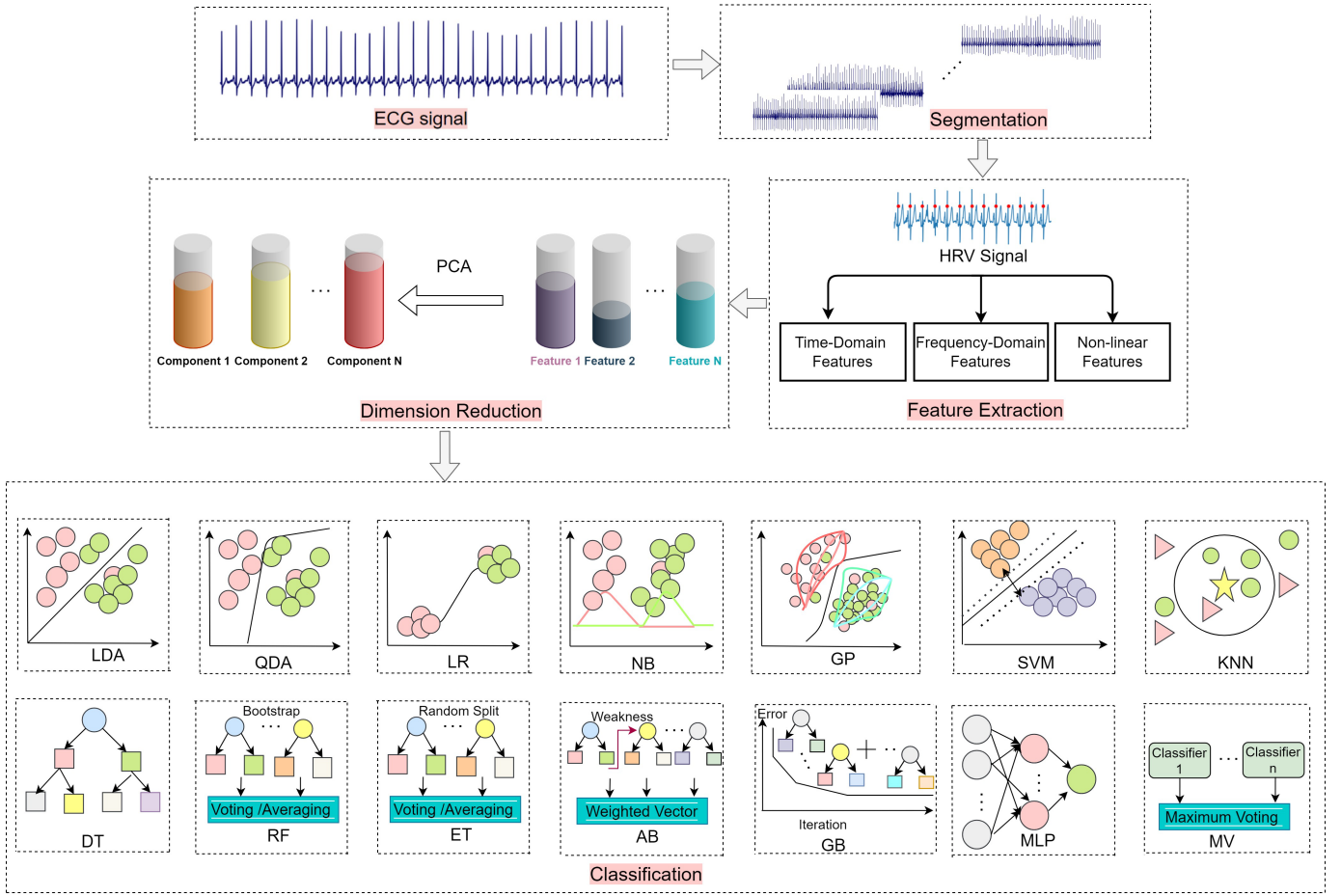


Fig. 1. Flowchart representation of our developed framework for the evaluation of conventional machine learning algorithms in the detection of sleep apnea from single-lead ECG.

the most effective and important features for classification. Here, we applied PCA for dimension reduction [19].

C. Classification

Different machine learning algorithms have been applied for sleep apnea detection. However, the main problem in this field is the lack of a fair and unified comparison between different algorithms. In this study, different well-known machine learning algorithms are compared. More information about these machine learning algorithms is available in [40].

1) *Linear Discriminant Analysis (LDA)*: LDA is a simple classifier that uses a linear decision boundary, generated by fitting Gaussian densities to each class using Bayes' rule [16]. The main hyperparameters that need to be predefined before training LDA are the solver algorithm and a tolerance parameter that defines the absolute threshold for a singular value to be considered significant.

2) *Quadratic Discriminant Analysis (QDA)*: QDA is similar to LDA but uses a quadratic boundary decision [17]. The solver for QDA is usually set to SVD, and therefore tolerance is the main hyperparameter that needs to be defined before training.

3) *Logistic Regression (LR)*: LR uses a logistic function for modeling dependent binary variables [20]. Solver algorithm,

tolerance for stopping criteria, regularization strength ($1/C$), and penalty norm are its main hyperparameters.

4) *Gaussian Naïve Bayes (GNB)*: GNB is a probabilistic method based on applying Bayes' theorem with naïve independence assumption between features. Its main hyperparameter is the portion of the largest variance of all features that has to be added to variances for stability [41].

5) *Gaussian Process (GP)*: GP classification is a nonparametric approach that uses a kernel function to assign class labels. Kernel, optimizer algorithm, and the maximum number of iterations for approximation posterior in Newton's method are its main hyperparameters.

6) *SVM*: SVM uses a kernel function for mapping data into a higher dimension. Lagrange equations are used to solve the problem in higher dimensions [21]. The type of the kernel and the kernel parameters are its main hyperparameters.

7) *K-Nearest Neighborhood (KNN)*: KNN classifies data samples based on their nearest neighbors. There are different criteria for the distance metric in the KNN method, including Euclidean distance and Mahalanobis distance [22]. Distance metric, the metric parameters, number of nearest neighbors, and leaf size are its main hyperparameters.

8) *Decision Tree (DT)*: DT is a nonparametric classifier that predicts the class labels by learning simple decision rules

inferred from the extracted features [18]. Interpretability is the main advantage of this method. Max depth of trees, the strategy used to choose the split, and the criterion measuring the quality of splits are the main hyperparameters of DT.

9) *Random Forest (RF)*: RF is an extended version of DT. RF uses several DTs for combinational learning by subsampling the input data with replacement (bootstrapping) [20]. This method largely prevents over-fitting during training, although improving performance in this method and reducing over-fitting depends on the type of dataset. Maximum depth, the minimum number of samples required to split an internal node, split criterion, and the number of estimators (trees) are the main hyperparameters to be set.

10) *Extra Tree (ET)*: ET is another evolved model based on DTs that is similar to RF but unlike RF that uses bootstrapping methods, uses the whole original sample. In addition, unlike RF which chooses the optimum cut points to split nodes, ET chooses them randomly. In terms of computational complexity and time required to implement the algorithms, ET is faster and more efficient than RF. Maximum depth, the minimum number of samples required to split an internal node, split criterion, and the number of estimators (trees) are the main hyperparameters to be set.

11) *AdaBoost (AB)*: AB combines several weak learners (decision stumps, trees with one node and two leaves) to build a stronger classifier. Unlike RF that samples the training data based on the bagging technique, AB, at each iteration, upweights the misclassified samples so that they appear in the training subset of the next learner with higher probability. After each learner is trained, a weight is assigned to it based on its performance. Number of estimators (trees), boosting algorithm, and learning rate are its main hyperparameters.

12) *Gradient Boosting (GB)*: GB is a generalization of AB that casts boosting as an optimization problem by using a gradient descent like procedure to minimize the loss of the model when adding new weak learners (here, DTs). Loss function, number of estimators, maximum depth, learning rate, the fraction of samples to be used for fitting the individual base learners, and split criterion are its main hyperparameters.

13) *Multi-Layer Perceptron (MLP)*: MLP is a feed-forward neural network consisting of multiple layers of nonlinear nodes capable of modeling complex and nonlinear relationships [20]. Its main hyperparameters are the number of layers, number of nodes, learning rate, type of activation functions used in the nodes, optimization method, and the loss tolerance to stop training. The number of hidden layers was set to 1.

14) *Majority Vote (MV)*: MV is an ensemble method for classification by combining several classifiers using a majority voting scenario. Here, we used MV to combine all the conventional machine learning algorithms.

V. DEEP LEARNING

Different CNNs and DRNNs [42] were employed to build standalone and hybrid deep models for the detection of sleep apnea. The flowchart of the proposed approach for sleep apnea detection using deep learning is shown in Fig. 2.

CNN provides a unique opportunity for automatic feature extraction and classification of images, signals, and other

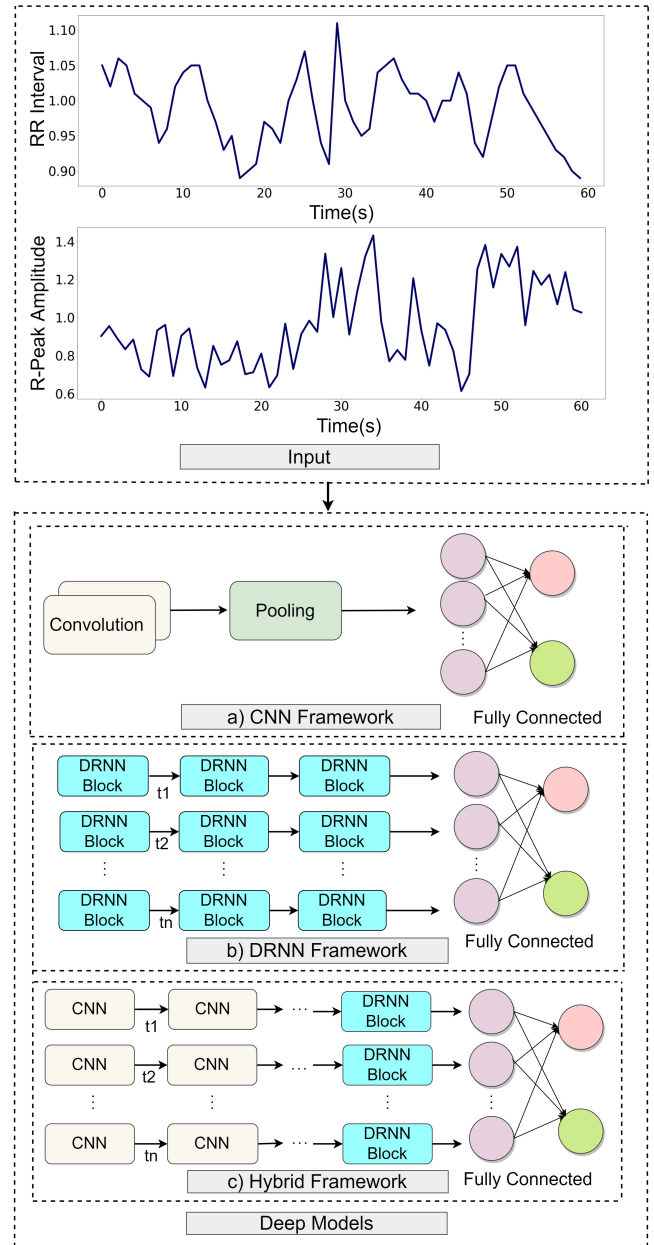


Fig. 2. Deep neural network architectures developed for the detection sleep apnea: (a) CNN, (b) DRNN, and (c) Hybrid CNN-DRNN framework.

biological data. VGG16, VGG19, ZF-Net, and Alex-Net are well-known CNN architectures that have led to numerous successful applications in image processing. Here, we similarly modified these networks, based on the dimension of our input data, to be applied in the detection of sleep apnea.

A. ZF-Net

The developed ZF-Net comprised of 96 kernels of size 7×1 stacked to a 3×1 max-pooling followed by a batch normalization in the first layer, a convolution layer with 256 kernels of size 5×1 followed by a 3×1 max-pooling and batch normalization in the second layer, and three convolution layers with 512 kernels of size 3×1 , 1024 kernels of size 3×1 , and 512 kernels of size 3×1 stacked together followed

by a 3×1 max-pooling in the third layer. Data were then flattened and fed to two fully connected layers with 418 and two nodes.

B. Alex-Net

The developed Alex-Net comprised of a convolution layer with 96 kernels of size 11×1 stacked to a batch normalization layer followed by a 3×1 max-pooling operation in the first layer, a convolution layer with 256 kernels of size 5×1 stacked to a batch normalization layer followed by a 3×1 max-pooling operation in the second layer, and two convolution layers with 384 kernels of size 3×1 , and a convolution layer with 256 kernels of size 3×1 each stacked to a batch normalization layer followed by a 3×1 max-pooling in the third layer. Data were then flattened and fed to two fully connected layers with 209 and two nodes.

C. VGG16

The developed VGG16 comprised of two convolution layers with 64 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the first layer, two convolution layers with 128 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the second layer, two convolution layers with 256 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the third layer, and two convolution layers with 512 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the fourth layer. Data were then flattened and fed to two fully connected layers with 512 and two nodes.

D. VGG19

The developed VGG19 comprised of two convolution layers with 64 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the first layer, two convolution layers with 128 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the second layer, four convolution layers with 256 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the third layer, four convolution layers with 512 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the fourth layer, and a convolution layer with 512 kernels of size 3×1 in the fifth layer. Data were then flattened and fed to two fully connected layers with 167 and two nodes.

E. LSTM

The developed LSTM comprised of a three-layer two-cell stacked architecture where each cell received a 2D structure of 90×2 . The output dimension was set to 13×2 . Data were then flattened and fed to two fully connected layers with four and two nodes.

F. BiLSTM

The developed BiLSTM comprised of a three-layer two-cell stacked architecture where each cell received a 2-D structure

of 90×2 . The output dimension was set to 26×2 . Data were then flattened and fed to two fully connected layers with four and two nodes.

G. GRU

The developed GRU comprised of three-layer two-cell stacked architecture where each cell received a 2-D structure of 90×2 . The output dimension was set to 22×2 . Data were then flattened and fed to two fully connected layers with seven and two nodes.

H. Hybrid ZFNet

The developed hybrid ZF-Net comprised of 96 kernels of size 7×1 stacked to a 3×1 max-pooling followed by a batch normalization in the first layer, 256 kernels of size 5×1 followed by a 3×1 max-pooling and batch normalization in the second layer, and 512 kernels of size 3×1 , 1024 kernels of size 3×1 , and 512 kernels of size 3×1 stacked together followed by a 3×1 max-pooling in the third layer. The output was then connected to a one-layer two-cell stacked DRNN (LSTM, BiLSTM, or GRU) where each cell received a 2-D structure of 512×4 . The output dimension was set to 128×2 for the GRU and LSTM while it was set to 256×2 for the BiLSTM. Data were then flattened and fed to two fully connected layers with 37 and two nodes.

I. Hybrid AlexNet

The developed hybrid Alex-Net comprised of 96 kernels of size 11×1 stacked to a batch normalization layer followed by a 3×1 max-pooling operation in the first layer, 256 kernels of size 5×1 stacked to a batch normalization layer followed by a 3×1 max-pooling in the second layer, and two convolution layers with 384 kernels of size 3×1 , and a convolution layer with 256 kernels of size 3×1 each stacked to a batch normalization layer followed by a 3×1 max-pooling in the third layer. The output was then connected to a one-layer two-cell stacked DRNN (LSTM, BiLSTM, or GRU) where each cell received a 2-D structure of 256×4 . The output dimension was set to 64×2 for the GRU and LSTM while it was 128×2 for the BiLSTM. Data were then flattened and fed to two fully connected layers with 18 and two nodes.

J. Hybrid VGG16

The developed hybrid VGG16 comprised of two convolution layers with 64 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the first layer, two convolution layers with 128 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the second layer, two convolution layers with 256 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the third layer, and two convolution layers with 512 kernels of size 3×1 stacked together followed by a 2×1 max-pooling operation in the fourth layer. The output was then connected to a one-layer two-cell stacked DRNN (LSTM, BiLSTM, or GRU) where each cell received a 2-D structure of 512×5 . The output dimension of the DRNN was set to

128×2 for the GRU and LSTM while it was 256×2 for the BiLSTM. Data were then flattened and fed to two fully connected layers with 37 and two nodes.

K. Hybrid VGG19

The developed VGG19 comprised of two convolution layers with 64 kernels of size 3×1 stacked together followed by a 2×1 max-pooling in the first layer, two convolution layers with 128 kernels of size 3×1 stacked together followed by a 2×1 max-pooling in the second layer, four convolution layers with 256 kernels of size 3×1 stacked together followed by a 2×1 max-pooling in the third layer, four convolution layers with 512 kernels of size 3×1 stacked together followed by a 2×1 max-pooling in the fourth layer, and a convolution layer with 512 kernels of size 3×1 were used in the fifth layer. The output was then connected to a one-layer two-cell stacked DRNN (LSTM, BiLSTM, or GRU) where each cell received a 2-D structure of 512×5 . The output dimension was set to 64×2 for the GRU and LSTM while it was 128×2 for BiLSTM. Data were then flattened and fed to two fully connected layers with 21 and two nodes.

VI. EXPERIMENTS AND DISCUSSION

We used PhysioNet Apnea-ECG Database v1.0.0 for apnea detection [32]. The dataset consists of 34 428 1-min intervals (normal: 21 362, apnea: 13 066) which means that it is imbalanced, that is, the number of apnea episodes is relatively lower than normal episodes. To overcome this problem, we used the stratified cross-validation method which is an effective method for balancing data in imbalanced datasets [43]. For a reliable and accurate analysis, we performed fivefold stratified cross-validation. Cross-validation allows us to train and test our models on multiple train-test splits. This provides a better indication of how well our model will perform on unseen data. In each fold, 70% of the data were used for training the models, 10% of the data were used as a validation set to find optimum hyperparameters (as reported in Sections IV and V), and 20% of data was used to evaluate the trained models on data not seen during training and validation.

Google Colaboratory Environment was used to implement machine learning and deep learning algorithms in Python. Keras and scikit-learn libraries were used for the implementation of deep learning and machine learning algorithms, respectively. hrvaranalysis, scipy, and entropy libraries were also used for feature extraction from ECG signals. An open-source version of the code developed for this manuscript has been made available on GitHub at <https://github.com/mahsaabahrmi/sleep-apnea>.

To find the most representative HRV features for the detection of sleep apnea, their performances were compared using the ET algorithm (Fig. 3). HF power was found as the most important feature for sleep apnea detection. This can be attributed to the fact that the HF component of HRV represents the respiration effect on heart rate as the result of the parasympathetic activity of the autonomous nervous system [35].

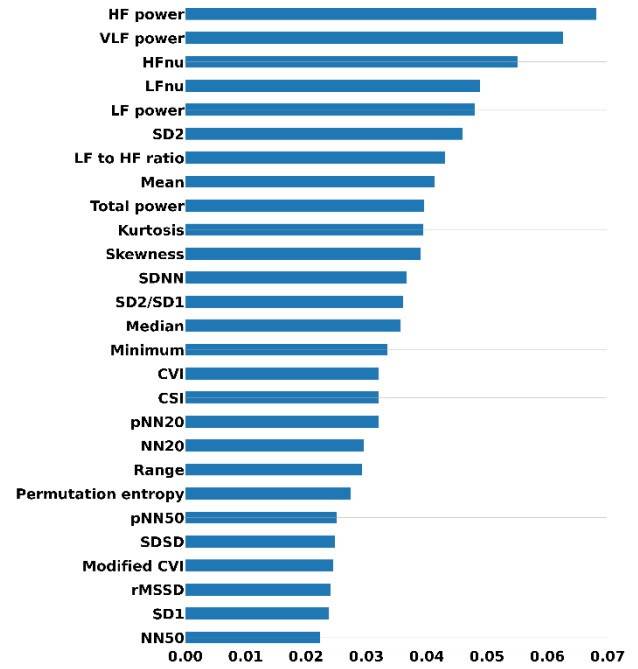


Fig. 3. HRV feature ranking for sleep apnea detection.

Followed by HF, other frequency bands including VLF, LF, and their normalized values had the most impact on the detection performance. The ratio of LF to HF and total power were other important features.

The HRV VLF component represents sympathovagal activity and has unique characteristics in cardiovascular disease diagnosis [44]. In addition, studies have shown that the VLF band has a strong correlation with the renin-angiotensin system, metabolic syndromes, and all-cause mortality after traumatic brain injury [44].

The LF component of HRV is considered as a marker of both sympathetic and parasympathetic modulation, thereby it is directly associated with breathing disorders [35].

Overall, the frequency-domain features were found more reliable and significant than other features in sleep apnea analysis, meaning that sleep apnea can be best detected by the analysis of HRV frequency bands.

Statistical features including mean, kurtosis, skewness, and SDNN were the most important time-domain features of HRV for the detection of apnea events.

Among nonlinear HRV features, SD2 was the most important feature. SD2 is extracted from the Poincaré plot, which is constructed by plotting each RR interval against the previous one. When analyzing the Poincaré plot by fitting an ellipse, SD2 means the standard deviation of the Poincaré plot along the line of identity [37]. SD2 measures short- and long-term HRV and is correlated with LF power and baroreflex sensitivity and therefore can represent the autonomous system activity and its malfunction.

Table I summarizes the performances of different conventional machine learning algorithms (on unseen test data) that rely on HRV feature extraction. The optimum model hyperparameters found on a validation set (separate from the test and train sets) are listed in the second column. The performances are reported in terms of accuracy, sensitivity,

TABLE I
PERFORMANCE COMPARISON BETWEEN DIFFERENT CONVENTIONAL MACHINE LEARNING ALGORITHMS IN SLEEP APNEA DETECTION

Model	Optimum Hyperparameters	Accuracy	Sensitivity	Specificity	F-Score
LDA	Solver = SVD, Tolerance = 10^{-4}	76.77%±0.51%	63.49%	84.93%	67.55%
QDA	Tolerance = 8.30×10^{-4}	75.54%±1.49%	74.70%	76.07%	69.95%
LR	Solver = 'LBFGS', Penalty = L2, C = 95.78, Tolerance = 10^{-4}	76.91%±0.55%	62.55%	85.73%	67.34%
GNB	Var_smoothing = 9.46×10^{-3}	75.96%±1.13%	68.16%	80.76%	68.36%
GP	Kernel = RBF, Max iteration = 3, Optimizer = 'LBFGS'	77.26%±0.38%	62.93%	86.08%	67.82%
SVM	Kernel = RBF, C = 2.52×10^2 , $\gamma = 0.75$	78.44%±0.24%	69.66%	83.83%	71.08%
KNN	Metric = Manhattan: P = 1, K = 11, Leaf size = 70	77.85%±0.75%	68.83%	83.40%	70.29%
DT	Max depth = 5, Strategy = Best split, Criterion = Gini	74.47%±0.78%	65.16%	80.20%	66.03%
RF	Min sample split = 2, Max depth = 11, Criterion = Gini, Number of estimators = 90	77.79%±0.60%	65.37%	85.44%	69.16%
ET	Min sample split = 2, Max depth = 28, Criterion = Gini, Number of estimators = 80	78.33%±0.31%	65.71%	86.08%	69.78%
AB	Number of estimators = 70, Boosting algorithm = 'SAMME.R', Learning rate = 0.39	77.09%±0.71%	64.32%	84.94%	68.13%
GB	Loss = Exponential, Criterion = MSE, Max depth = 24, Number of estimators = 80, Fraction = 0.59	77.52%±0.56%	64.79%	85.35%	68.68%
MLP	Learning rate = 10^{-3} , Activation function = Relu, Optimizer = Adam, Hidden neurons = 50, alpha = 10^{-4} , Tolerance = 10^{-4}	78.52%±0.52%	70.40%	83.53%	71.40%
MV	Best parameters of each model.	79.39%±0.65%	68.73%	85.94%	71.75%

specificity, and F -score. By combining all the classifiers through an MV scheme, MV achieved the best accuracy (79.39%) and F -score (71.75%). It was observed that QDA achieves the best sensitivity (74.70%) while ET and GP achieve the best specificity (86.08%). We conjecture that a more complex fusion method that can weigh the classifiers' output based on their performance and reliability may further improve the detection performance [45].

Among individual classifiers, MLP and SVM achieved superior results. Thanks to its powerful nonlinear modeling capability through the complex interactions of several nonlinear nodes, MLP can extract nonlinear input-output relationships. In fact, it has been theoretically proved that an MLP with a hidden layer and enough hidden nodes is capable of modeling any nonlinear function with a finite number of discontinuities [46]. SVM is a highly regularized classifier that performs well when there is not enough data available [47].

Among tree-based algorithms, ET and RF achieved superior results. Compared to DT which gives high importance to a particular set of features, RF and ET select features randomly during the training process. Therefore, they do not depend on any specific set of features and can better generalize. Compared to GB and AB which are boosting algorithms, RF is an ensemble bagging algorithm. By generating new data using combinations with repetitions, bagging decreases the variance in the prediction and is less prone to overfitting.

Among different conventional machine learning methods, LDA, QDA, DT, GNB, LR, GP, and AB had the lowest accuracy. Given the simplicity of such algorithms, their accuracy may degrade in modeling complex physiological processes.

Table II summarizes the performances of different deep learning algorithms on the test data in terms of accuracy, sensitivity, specificity, and F -score. Among different methods, a hybrid model of ZFNet-BiLSTM achieved the best accuracy (88.13%) and specificity (92.27%), ZFNet-GRU achieved the

best sensitivity (84.26%), and VGG16-LSTM achieved the best F -score (84.24%).

It was found that deep neural networks were superior to their conventional machine learning counterparts because they have the power of extracting the most important features, automatically. Comparing the best deep model (ZFNet-BiLSTM) with the best machine learning model (MV) using a paired t -test, it was revealed that the deep model's accuracy is significantly higher than the best conventional machine learning model ($p < 0.01$).

While DRNNs (LSTM, BiLSTM, and GRU) performed better than conventional machine learning methods, their performance was lower than CNN-based networks (ZFNet, AlexNet, VGG16, and VGG19). This can be attributed to the fact that our methods were based on the analysis of short 1-min ECG segments. We conjecture that if the apnea detection is performed on longer ECG segments, the performance of DRNNs will be improved as these networks are particularly powerful in the analysis of long time-series data.

Among different CNNs, VGG16, ZFNet, and AlexNet achieved slightly better performance compared to VGG19. VGG19 is a deeper network than other CNNs implemented in this article. However, we had to modify the depth of this network based on our dataset. We conjecture that the modified structure of the VGG19 affected its ability to extract the most effective features and consequently slightly lower classification performance.

Considering that a lower rate of false negative is of higher importance in the detection of apnea, the sensitivity of the designed classifiers was studied. Among different conventional classifiers, QDA, MLP, SVM, KNN, and MV had the best sensitivities. Most deep learning algorithms outperformed the conventional machine learning algorithms with respect to sensitivity where ZFNet-GRU and AlexNet-GRU achieved the highest performance.

TABLE II
PERFORMANCE COMPARISON BETWEEN DIFFERENT DEEP
LEARNING ALGORITHMS IN SLEEP APNEA DETECTION

Model	Accuracy	Sensitivity	Specificity	F-Score
ZFNet	87.36%±0.25%	81.64%	90.92%	83.20%
AlexNet	87.09%±0.89%	82.64%	89.87%	83.10%
VGG16	87.26%±0.29%	82.06%	90.50%	83.18%
VGG19	86.75%±0.38%	81.44%	90.07%	82.52%
LSTM	82.52%±0.29%	73.93%	87.88%	76.46%
BiLSTM	82.45%±0.20%	73.14%	88.25%	76.17%
GRU	82.93%±0.40%	73.80%	88.63%	76.86%
VGG16-LSTM	88.02%±0.42%	83.34%	90.95%	84.24%
VGG16-GRU	87.78%±0.67%	83.35%	90.53%	83.97%
VGG16-BiLSTM	88.01%±0.34%	82.90%	91.20%	84.15%
VGG19-LSTM	87.06%±0.49%	82.06%	90.17%	82.96%
VGG19-GRU	86.62%±0.65%	81.32%	89.93%	82.36%
VGG19-BiLSTM	86.92%±0.31%	81.53%	90.28%	82.72%
AlexNet-LSTM	87.32%±0.33%	81.10%	91.20%	83.06%
AlexNet-GRU	87.11%±0.48%	84.12%	88.98%	83.37%
AlexNet-BiLSTM	87.43%±0.15%	82.67%	90.40%	83.46%
ZFNet-LSTM	87.84%±0.42%	83.29%	90.68%	84.03%
ZFNet-GRU	87.43%±0.28%	84.26%	89.42%	83.74%
ZFNet-BiLSTM	88.13%±0.26%	81.49%	92.27%	84.04%

Hybrid CNN-DRNN architectures benefit from the power of CNNs in feature extraction and the power of DRNNs in modeling the time pattern of the extracted features over time. In this scenario, the CNN network extracts high-resolution features while the DRNN models the evolution of these features over a coarser time interval. Therefore, it is expected that they achieve higher performance [48]. Among different hybrid networks, ZFNet-BiLSTM outperformed the rest. ZFNet is a modified version of AlexNet by reducing the filter size that leads to better accuracy. Combining a BiLSTM to capture the time pattern of the features extracted by ZFNet led to the highest performance in the detection of sleep apnea.

Our developed deep learning framework was based on the analysis of ECG R-R intervals and R-peak amplitudes. To study the contribution of R-R intervals and R-peak amplitude, our best model (ZFNet-BiLSTM) was once implemented using only the R-R intervals as the input and once using the R-peak amplitudes as the input. We found that the accuracy of our best model is dropped to $84.62\% \pm 0.51\%$ ($\sim 4\%$ decrease) by feeding solely the R-R intervals and to $83.59\% \pm 0.53\%$ ($\sim 5\%$ decrease) by feeding solely the R-peak amplitudes to the model. This shows that the R-R intervals are more informative and have more predictive value compared to the R-peak amplitudes for sleep apnea detection. The combination of R-R intervals and R-peak amplitudes achieves the best performance.

Many methods have been proposed recently for the detection of sleep apnea from different physiological signals. In this article, we compared our models with some relatively new methods that are based on the analysis of single-lead ECG using the same dataset as ours [32]. Table III compares the performance of our best model with state-of-the-art sleep apnea detection methods in terms of accuracy, sensitivity, and specificity. It can be observed that our reported results are superior to those reported in [1], [2], [7], [9], [13], and [23], closely match those reported in [4] and [11], and are inferior to [5]. It should be noted that unlike most studies that used only 35 recordings [1], [2], [4], [5], [7], [11], [13], [23], to achieve a more comprehensive evaluation, in this article we analyzed all the 70 recordings using a fivefold cross-validation technique. Our experiments were then repeated ten times and the results were averaged for a more reliable assessment. Moreover, unlike most studies that did not use a separate set of data to find the optimum hyperparameters of the models [2], [4], [5], [7], [9], [11], [13], [23], in this study each fold was divided into 70% training, 10% validation (to tune the hyperparameters and avoid overfitting), and 20% of data to test the performance. All the reported results in the article are based on the test data not seen during training and validation. Obviously, if the optimum hyperparameters are chosen based on the test data, the achieved results are not generalizable [49]. The limitations of the reviewed studies are listed in the last column of Table III.

Many of the existing apnea detection algorithms are based on the analysis of several minutes of data before and after the occurrence of apnea limiting their applications in real-time detection and monitoring of apnea events. Here, we limited our analysis to the examination of 1-min segments of ECG signal to detect whether they contain an apnea event. Therefore, our algorithm can detect apnea events with a time resolution of 1 min. Future work should be directed toward the design and evaluation of machine learning and deep learning algorithms to predict (forecast) sleep apnea events in the future [54]. Here, we performed a new experiment to forecast the occurrence of sleep apnea events in the future using our best model (ZFNet-BiLSTM). Our model achieved an accuracy, sensitivity, specificity, and F-score of $84.69\% \pm 0.34\%$, 77.10% , 89.45% , and 79.46% in the prediction of apnea events 1 min in future, $83.54\% \pm 0.31\%$, 78.33% , 86.81% , and 78.57% in the prediction of apnea events 2 min in future, and $80.23\% \pm 0.34\%$, 73.45% , 84.52% , and 74.18% in the prediction of apnea events 5 min in future. This shows the capability of our design algorithms in accurate detection as well as prediction of apnea events. Further research should be performed to comprehensively evaluate and compare the performance of different algorithms in the prediction of apnea events ahead of time.

Noise and motion artifacts are major problems in the analysis of physiological signals. Here, we simply filtered the ECG signals using a median filter. To further improve the reliability of apnea detection, the use of artifact rejection and correction methods is recommended where those portions of the data that are highly corrupted are excluded from the analysis [50] or corrected based on autoregressive models [51].

TABLE III

COMPARISON OF DIFFERENT STATE-OF-THE-ART ALGORITHMS FOR THE DETECTION OF SLEEP APNEA ON APNEA-ECG DATABASE. UNLIKE MOST RECENT PUBLICATIONS, OUR DEVELOPED MODELS WERE TRAINED AND TESTED ON THE WHOLE 70 RECORDINGS USING A FIVEFOLD CROSS VALIDATION. WITHIN EACH FOLD, 10% OF THE TRAINING DATASET WAS USED AS A VALIDATION SET TO SET THE MODELS' HYPERMETERS WHILE REPORTING THE RESULTS ON UNSEEN TEST DATA. OUR EXPERIMENTS WERE PERFORMED TEN TIMES AND THE AVERAGE PERFORMANCE WAS REPORTED FOR A MORE RELIABLE EVALUATION

Model	Accuracy	Sensitivity	Specificity	Limitations
LeNet-LSTM [1]	80.67%	75.04%	84.13%	The classifier was tested only on 35 records of the 70 recordings.
ANN [2]	82.12%	88.41%	72.29%	The classifier was trained and tested on only 35 records of the 70 recordings. Selection of the optimum model hyperparameters was not discussed.
Scalogram [4]	86.22%	90.00%	83.82%	The classifier was trained and tested on only 35 records of the 70 recordings. Selection of the optimum model hyperparameters was not discussed.
CNN [5]	98.91%	98.72%	99.20%	Only 50% of 35 recordings were used to train the classifier and the rest was used to test it. Selection of the optimum model hyperparameters was not discussed.
LSTM [7]	81.30%	59.90 %	91.75%	The classifier was only tested on 35 records of the 70 recordings
Unsupervised feature learning [9]	80.70%	80.10%	81.00%	Selection of the optimum model hyperparameters was not discussed.
MSDA-1DCNN+WLTD [11]	89.40%	89.80%	89.10%	The classifier was only tested on 35 records of the 70 recordings Selection of model hyperparameters was performed based on the same train and test data.
Wise feature selection [13]	82.12%	88.41%	72.29%	Trained and tested on 35 recordings using a 10-fold cross-validation. Selection of model hyperparameters was performed based on the same train and test data.
Hidden Markov model [23]	84.70%	88.90%	82.10%	The classifier was only tested on 35 records of the 70 recordings Selection of model hyperparameters was performed based on the same train and test data.
Our best model	88.13%	81.49%	92.27%	--

Our study was performed on PhysioNet Apnea-ECG Database v1.0.0. The small number of mild and moderate OSA patients and the lack of CSA episodes are the main limitations of this database that needs to be addressed in future work by collecting a larger set of data with different types of apneas.

VII. CONCLUSION

The advancement of smart wearables technologies has provided a unique opportunity for sleep and health monitoring. However, wearable technologies rely on accurate and real-time monitoring algorithms. In this article, we provided a comprehensive comparison between different machine learning and deep learning algorithms, in a unified framework, for the detection of sleep apnea from single-lead ECG. It was observed that deep learning models outperform conventional machine learning techniques. Among deep learning algorithms, CNN-based models performed better than DRNNs in our application where short segments of ECG (1 min) were processed. The best detection performances were obtained using hybrid CNN-DRNN architectures among which ZFNet-BiLSTM achieved the highest accuracy and specificity and ZFNet-GRU achieved the best sensitivity. Based on the achieved results, the use of hybrid deep neural networks is recommended for sleep apnea detection from ECG.

This work provides valuable information to sleep researchers on the design and selection of appropriate machine learning and deep learning algorithms for the detection of sleep apnea. The algorithms can also be extended toward the detection and monitoring of other sleep events, such as hot flashes and arousals by analyzing the physiological signals associated with these events [52], [53].

REFERENCES

- [1] M. Bahrami and M. Forouzanfar, "Detection of sleep apnea from single-lead ECG: Comparison of deep learning algorithms," in *Proc. IEEE Int. Symp. Med. Meas. Appl. (MeMeA)*, Jun. 2021, pp. 1–5.
- [2] N. Pombo, B. M. C. Silva, A. M. Pinho, and N. Garcia, "Classifier precision analysis for sleep apnea detection using ECG signals," *IEEE Access*, vol. 8, pp. 200477–200485, 2020.
- [3] M. H. Kryger, T. Roth, and W. C. Dement, *Principles and Practice of Sleep Medicine*, 6th ed. Amsterdam, The Netherlands: Elsevier, 2017.
- [4] S. A. Singh and S. Majumder, "A novel approach osa detection using single-lead ECG scalogram based on deep neural network," *J. Mech. Med. Biol.*, vol. 19, no. 4, Jun. 2019, Art. no. 1950026.
- [5] D. Dey, S. Chaudhuri, and S. Munshi, "Obstructive sleep apnoea detection using convolutional neural network based deep learning framework," *Biomed. Eng. Lett.*, vol. 8, no. 1, pp. 95–100, Feb. 2018.
- [6] J. Acquavella, R. Mehra, M. Bron, J. M.-H. Suomi, and G. P. Hess, "Prevalence of narcolepsy and other sleep disorders and frequency of diagnostic tests from 2013–2016 in insured patients actively seeking care," *J. Clin. Sleep Med.*, vol. 16, no. 8, pp. 1255–1263, Aug. 2020.
- [7] O. Faust, R. Barika, A. Shenfield, E. J. Ciccio, and U. R. Acharya, "Accurate detection of sleep apnea with long short-term memory network based on RR interval signals," *Knowl.-Based Syst.*, vol. 212, Jan. 2021, Art. no. 106591.
- [8] B. M. Altevogt and H. R. Colten, *Sleep Disorders and Sleep Deprivation: An Unmet Public Health Problem*. Washington, DC, USA: National Academies Press (U.S.), 2006.
- [9] K. Feng, H. Qin, S. Wu, W. Pan, and G. Liu, "A sleep apnea detection method based on unsupervised feature learning and single-lead electrocardiogram," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.
- [10] A. B. Neikrug and S. Ancoli-Israel, "Sleep disorders in the older adult—A mini-review," *Gerontology*, vol. 56, no. 2, pp. 181–189, 2010.
- [11] Q. Shen, H. Qin, K. Wei, and G. Liu, "Multiscale deep neural network for obstructive sleep apnea detection using RR interval from single-lead ECG signal," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.
- [12] M. M. Lyons, N. Y. Bhatt, A. I. Pack, and U. J. Magalang, "Global burden of sleep-disordered breathing and its implications," *Respirology*, vol. 25, no. 7, pp. 690–702, Jul. 2020.
- [13] A. Pinho, N. Pombo, B. M. C. Silva, K. Bousson, and N. Garcia, "Towards an accurate sleep apnea detection based on ECG signal: The quintessential of a wise feature selection," *Appl. Soft Comput.*, vol. 83, Oct. 2019, Art. no. 105568.

- [14] W. Conwell *et al.*, "Prevalence, clinical features, and CPAP adherence in REM-related sleep-disordered breathing: A cross-sectional analysis of a large clinical population," *Sleep Breathing*, vol. 16, no. 2, pp. 519–526, Jun. 2012.
- [15] B. Fatimah, P. Singh, A. Singhal, and R. B. Pachori, "Detection of apnea events from ECG segments using Fourier decomposition method," *Biomed. Signal Process. Control*, vol. 61, Aug. 2020, Art. no. 102005.
- [16] J. V. Marcos, R. Hornero, D. Álvarez, F. D. Campo, and M. Aboy, "Automated detection of obstructive sleep apnoea syndrome from oxygen saturation recordings using linear discriminant analysis," *Med. Biol. Eng. Comput.*, vol. 48, no. 8, pp. 895–902, Sep. 2010.
- [17] J. V. Marcos, R. Hornero, D. Álvarez, F. del Campo, and C. Zamarrón, "Assessment of four statistical pattern recognition techniques to assist in obstructive sleep apnoea diagnosis from nocturnal oximetry," *Med. Eng. Phys.*, vol. 31, no. 8, pp. 971–978, Oct. 2009.
- [18] H. Ting, Y.-T. Mai, H.-C. Hsu, H.-C. Wu, and M.-H. Tseng, "Decision tree based diagnostic system for moderate to severe obstructive sleep apnea," *J. Med. Syst.*, vol. 38, no. 9, pp. 1–10, Sep. 2014.
- [19] S. M. Isa, M. I. Fanany, S. Jatmiko, and A. M. Arymurthy, "Sleep apnea detection from ECG signal: Analysis on optimal features, principal components, and nonlinearity," in *Proc. 5th Int. Conf. Bioinf. Biomed. Eng.*, May 2011, pp. 1–4.
- [20] A. Bostanci, M. Turhan, and S. Bozkurt, "Can statistical machine learning algorithms help for classification of obstructive sleep apnea severity to optimal utilization of polysomnography resources?" *Methods Inf. Med.*, vol. 56, no. 4, pp. 308–318, 2017.
- [21] V. Karunakaran, M. Saran, S. Abishek, G. V. TG, A. Vojaswain, and K. Ramachandran, "Detection of obstructive sleep apnea from ECG signal using SVM based grid search," *Int. J. Electron. Telecommun.*, vol. 67, no. 1, pp. 5–12, 2021.
- [22] M. Sharma, M. Raval, and U. R. Acharya, "A new approach to identify obstructive sleep apnea using an optimal orthogonal wavelet filter bank with ECG signals," *Informat. Med. Unlocked*, vol. 16, 2019, Art. no. 100170.
- [23] K. Li, W. Pan, Y. Li, Q. Jiang, and G. Liu, "A method to detect sleep apnea based on deep neural network and hidden Markov model using single-lead ECG signal," *Neurocomputing*, vol. 294, pp. 94–101, Jun. 2018.
- [24] B. Amra, B. Rahmati, F. Soltaninejad, and A. Feizi, "Screening questionnaires for obstructive sleep apnea: An updated systematic review," *Oman Med. J.*, vol. 33, no. 3, p. 184, 2018.
- [25] S. Tsuike *et al.*, "Machine learning for image-based detection of patients with obstructive sleep apnea: An exploratory study," *Sleep Breathing*, vol. 25, no. 4, pp. 2297–2305, 2021.
- [26] M. A. Secretariat, "Polysomnography in patients with obstructive sleep apnea: An evidence-based analysis," *Ontario Health Technol. Assessment*, vol. 6, no. 13, p. 1, 2006.
- [27] M. Mabrouk, S. Rajan, M. Bolic, M. Forouzanfar, H. R. Dajani, and I. Batkin, "Human breathing rate estimation from radar returns using harmonically related filters," *J. Sensors*, vol. 2016, pp. 1–7, Jan. 2016.
- [28] M. De Zambotti, N. Cellini, A. Goldstone, I. M. Colrain, and F. C. Baker, "Wearable sleep technology in clinical and research settings," *Med. Sci. Sports Exerc.*, vol. 51, no. 7, p. 1538, 2019.
- [29] Y. S. Lee, P. N. Pathirana, R. J. Evans, and C. L. Steinfort, "Noncontact detection and analysis of respiratory function using microwave Doppler radar," *J. Sensors*, vol. 2015, pp. 1–13, Jan. 2015.
- [30] G. Surrel, A. Aminifar, F. Rincón, S. Murali, and D. Atienza, "Online obstructive sleep apnea detection on medical wearable sensors," *IEEE Trans. Biomed. Circuits Syst.*, vol. 12, no. 4, pp. 762–773, Aug. 2018.
- [31] M. Forouzanfar, M. Mabrouk, S. Rajan, M. Bolic, H. R. Dajani, and V. Z. Groza, "Event recognition for contactless activity monitoring using phase-modulated continuous wave radar," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 2, pp. 479–491, Feb. 2017.
- [32] T. Penzel, G. B. Moody, R. G. Mark, A. L. Goldberger, and J. H. Peter, "The apnea-ECG database," in *Proc. Comput. Cardiol.*, vol. 27, Sep. 2000, pp. 255–258.
- [33] P. Hamilton, "Open source ECG analysis," in *Proc. Comput. Cardiol.*, 2002, pp. 101–104.
- [34] L. Chen, X. Zhang, and C. Song, "An automatic screening approach for obstructive sleep apnea diagnosis based on single-lead electrocardiogram," *IEEE Trans. Autom. Sci. Eng.*, vol. 12, no. 1, pp. 106–115, Jan. 2015.
- [35] V. C. C. Sequeira, P. M. Bandeira, and J. C. M. Azevedo, "Heart rate variability in adults with obstructive sleep apnea: A systematic review," *Sleep Sci.*, vol. 12, no. 3, p. 214, 2019.
- [36] F. Shaffer and J. P. Ginsberg, "An overview of heart rate variability metrics and norms," *Frontiers Public Health*, vol. 5, p. 258, Sep. 2017.
- [37] Ş. C. Mirescu and S. Harden, "Nonlinear dynamics methods for assessing heart rate variability in patients with recent myocardial infarction," *Rom. J. Biophys.*, vol. 22, no. 2, pp. 117–124, 2012.
- [38] J. Jeppesen, S. Beniczky, P. Johansen, P. Sidenius, and A. Fuglsang-Frederiksen, "Using Lorenz plot and cardiac sympathetic index of heart rate variability for detecting seizures for patients with epilepsy," in *Proc. 36th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2014, pp. 4563–4566.
- [39] C. Bandt and B. Pompe, "Permutation entropy: A natural complexity measure for time series," *Phys. Rev. Lett.*, vol. 88, no. 17, Apr. 2002, Art. no. 174102.
- [40] M. Mohammed, M. B. Khan, and E. B. M. Bashier, *Machine Learning: Algorithms and Applications*. Boca Raton, FL, USA: CRC Press, 2016.
- [41] N. A. Eisman, M. B. Westover, J. E. Miettus, R. J. Thomas, and M. T. Bianchi, "Classification algorithms for predicting sleepiness and sleep apnea severity," *J. Sleep Res.*, vol. 21, no. 1, pp. 101–112, Feb. 2012.
- [42] Y. Bengio, I. Goodfellow, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2017.
- [43] X. Zeng and T. R. Martinez, "Distribution-balanced stratified cross-validation for accuracy estimation," *J. Experim. Theor. Artif. Intell.*, vol. 12, no. 1, pp. 1–12, 2000.
- [44] H. Usui and Y. Nishida, "The very low-frequency band of heart rate variability represents the slow recovery component after a mental stress task," *PLoS ONE*, vol. 12, no. 8, Aug. 2017, Art. no. e0182611.
- [45] M. Forouzanfar, H. R. Dajani, V. Z. Groza, M. Bolic, S. Rajan, and I. Batkin, "Bayesian fusion algorithm for improved oscillometric blood pressure estimation," *Med. Eng. Phys.*, vol. 38, no. 11, pp. 1300–1304, Nov. 2016.
- [46] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Math. Control, Signals Syst.*, vol. 2, no. 4, pp. 303–314, 1989.
- [47] I. Ahmad, M. Bashari, M. J. Iqbal, and A. Raheem, "Performance comparison of support vector machine, random forest, and extreme learning machine for intrusion detection," *IEEE Access*, vol. 6, pp. 33789–33795, 2018.
- [48] Q. Yao, R. Wang, X. Fan, J. Liu, and Y. Li, "Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network," *Inf. Fusion*, vol. 53, pp. 174–182, Jan. 2020.
- [49] K. Jankowsky and U. Schroeders, "Validation and generalizability of machine learning prediction models on attrition in longitudinal studies," *Int. J. Behav. Develop.*, pp. 1–8, 2021, doi: 10.1177/01650254221075034.
- [50] M. Forouzanfar, F. C. Baker, I. M. Colrain, A. Goldstone, and M. Zambotti, "Automatic analysis of pre-ejection period during sleep using impedance cardiogram," *Psychophysiology*, vol. 56, no. 7, Jul. 2019, Art. no. e13355.
- [51] M. Forouzanfar, F. C. Baker, M. de Zambotti, C. McCall, L. Giovannardi, and G. T. A. Kovacs, "Toward a better noninvasive assessment of preejection period: A novel automatic algorithm for B-point detection and correction on thoracic impedance cardiogram," *Psychophysiology*, vol. 55, no. 8, Aug. 2018, Art. no. e13072.
- [52] F. C. Baker *et al.*, "Changes in heart rate and blood pressure during nocturnal hot flashes associated with and without awakenings," *Sleep*, vol. 42, no. 11, p. zsz175, Oct. 2019.
- [53] H. Li and Y. Guan, "DeepSleep convolutional neural network allows accurate and fast detection of sleep arousal," *Commun. Biol.*, vol. 4, no. 1, pp. 1–11, Dec. 2021.
- [54] M. Bahrami and M. Forouzanfar, "Deep learning forecasts the occurrence of sleep apnea from single-lead ECG," *Cardiovascular Eng. Technol.*, 2022, doi: 10.1007/s13239-022-00615-50.